Title

Calibrated Conformal Generative Document Extraction: End-to-End OCR-Free Vision–Language Modeling with Structured Coverage Guarantees

Abstract

This work presents a fully generative, OCR-free multimodal vision–language model for structured document information extraction with formally calibrated uncertainty. Unlike prior approaches that bolt standard Monte Carlo dropout or heuristic confidence heads onto Donut-style architectures, the proposed framework embeds token- and field-level conformal prediction directly into a T5-style generative decoder, yielding coverage-guaranteed prediction sets over structured outputs (key–value pairs, tables, fields). A geometry-aware conformal scoring function leverages visual layout, reading order, and content embeddings to construct region-sensitive nonconformity scores, while a hierarchical calibration scheme enforces coverage at token-, field-, and document-level simultaneously. The model is trained end-to-end with a joint objective that couples sequence likelihood, conformal score regularization, and differentiable surrogate coverage losses. At inference, the model produces structured prediction sets and an abstention-aware extraction policy that optimizes expected utility under coverage constraints. Extensive experiments on revised, de-templated versions of FUNSD, CORD, SROIE, and multi-page DocVQA, with synthetic and real noise, demonstrate that the method matches or improves SOTA accuracy while significantly improving calibration, coverage, and selective risk. The design is optimized to be trainable on a single RTX 5050 GPU via parameter-efficient adaptation and conformal calibration atop small open-source backbones. The result is a TPAMI/ACL-grade framework that turns generative document extraction into a rigorously calibrated, uncertainty-aware prediction problem.

Core idea (2–3 lines)

End-to-end OCR-free generative document extraction with **hierarchical, geometry-aware conformal prediction** directly embedded in the decoder, producing token/field/document-level **prediction sets with provable coverage**. The model outputs not just values, but calibrated sets and abstentions, enabling trustworthy extraction under explicit risk–coverage tradeoffs, all trainable on a single consumer GPU via parameter-efficient tuning.

Research hypothesis

A conformal-prediction-integrated generative vision–language model with geometry-aware nonconformity scores and hierarchical coverage constraints can (1) match or exceed SOTA accuracy on document extraction benchmarks, and (2) significantly improve calibration, selective risk, and robustness to layout/noise compared to standard MC-dropout- or confidence-head-based uncertainty baselines, while remaining trainable on a single RTX 5050 GPU.

Key novelty contributions

1. **Hierarchical Conformal Generative Decoder**: A new decoder formulation that produces token-, field-, and document-level prediction sets for structured document outputs, with multi-level conformal coverage constraints baked into the decoding and training objectives.
2. **Geometry-Aware Nonconformity Scoring**: A multimodal nonconformity score that fuses visual layout, spatial structure, and semantic content to define document-specific conformal scores for key–value pairs and tables, rather than treating outputs as flat sequences.
3. **Differentiable Surrogate Coverage Loss**: A novel loss that encourages conformal scores and prediction set sizes to approximate desired coverage levels during training, enabling end-to-end optimization of calibration quality instead of post-hoc conformal wrapping only.

4. **Abstention-Aware Structured Decoding Policy**: A decoding algorithm that jointly chooses prediction sets and abstentions under a utility function (e.g., F1–coverage tradeoff), extending selective classification to fully generative structured outputs.
5. **Single-GPU-Efficient Design**: A parameter-efficient adaptation scheme (LoRA + layer-wise scaling + low-rank visual adapters) and calibration protocol explicitly engineered to fit within RTX 5050 constraints, demonstrating that rigorous uncertainty with conformal prediction is feasible on consumer hardware for document extraction.

Final architecture design

- **Backbone encoder**:
  - OCR-free vision encoder (small ViT/Swin) operating directly on document images (single and multi-page via page-wise encoding + learned page-position embeddings).
  - Optional shallow text encoder for weak supervision using pseudo-OCR tokens if available, fused via cross-attention but not required at inference (still OCR-free by default).
- **Generative decoder**:
  - T5-small/base-sized transformer decoder with:
    - Standard autoregressive generation over a structured serialization of document fields (JSON-like schema with slots for keys, values, table rows/columns).
    - Additional **conformal score head** at each decoding step producing per-token nonconformity logits.
    - Field-level aggregation modules that pool token nonconformity into field scores via attention over layout-aware token embeddings.
- **Geometry-aware fusion**:
  - For each token position, a fused representation concatenating: visual region features, positional/layout embeddings (absolute and relative 2D coordinates, block indices), and semantic context.
  - A small MLP projects the fused representation to nonconformity scores used by the conformal prediction layer.
- **Hierarchical conformal layer** (nonparametric component atop decoder outputs):
  - Uses calibration splits to compute quantiles of nonconformity at token, field, and document level.
  - Stores per-task, per-field quantiles used in decoding to define prediction sets and abstentions.

Learning objectives / loss functions
Let y be the structured output (sequence of tokens with field segmentation), $\hat{y}$ the model output, $s(\cdot)$ nonconformity scores, and τ the learned/target quantile thresholds.

1. **Generative loss (L_gen)**

- Standard negative log-likelihood over the serialized sequence:
  - $L_{gen} = -\log p_\theta(y \mid x)$
- Encourages accurate structured generation.

2. **Conformal score regularization (L_{conf})**
   - For each token i with nonconformity score s_i and empirical calibration quantile q_α (for target coverage $1-\alpha$):
     - Encourage scores to be well-separated: positives near or below q_α, incorrect/hard examples above q_α.
   - Implemented as a margin-based loss:
     - $L_{conf} = E[(s_i - q_\alpha + m)_+ \text{ for correct tokens}] + E[(q_\alpha - s_i + m)_+ \text{ for incorrect tokens}]$
   - m is a margin hyperparameter.
3. **Differentiable surrogate coverage loss (L_{cov})**
   - Approximate the coverage constraint by a soft indicator over whether true tokens/fields fall into the learned prediction set (defined by $s_i \le \hat{\tau}$, where $\hat{\tau}$ is a differentiable approximation of q_α).
   - L_{cov} penalizes empirical deviation of coverage from target $(1-\alpha)$ at multiple levels:
     - $L_{cov} = |\hat{C}_{token} - (1-\alpha_{token})| + |\hat{C}_{field} - (1-\alpha_{field})| + |\hat{C}_{doc} - (1-\alpha_{doc})|$
   - $\hat{C}_*$ are differentiable coverage surrogates using sigmoid relaxations.
4. **Regularization of set size (L_{size})**
   - Penalize overly large prediction sets:
     - $L_{size} = \lambda_{size} \cdot E[\#\text{candidates per token/field}]$
5. **Total loss**
   - $L_{total} = L_{gen} + \lambda_{conf} L_{conf} + \lambda_{cov} L_{cov} + \lambda_{size} L_{size}$
   - λ_{conf}, λ_{cov}, λ_{size} tuned via small grid search; ablations will test their impact.

Training strategy

- **Stage 1: Backbone warm-start**
  - Initialize encoder and decoder from compact OCR-free checkpoints (e.g., Donut-base or Pix2Struct-base) using HF models.
  - Freeze encoder (or apply small LoRA adapters) and fine-tune decoder on supervised document extraction without uncertainty modules (only L_{gen}).
- **Stage 2: Uncertainty module activation**

    - Unfreeze conformal score heads and geometry-aware fusion layers; keep most backbone weights frozen or LoRA-tuned to stay within RTX 5050 limits.
    - Train with L_total on training split; maintain a held-out calibration split.
- **Stage 3: Conformal calibration**
    - After training, fix θ and compute empirical nonconformity scores on calibration set.
    - Compute per-task, per-field quantiles qα for token/field/doc levels.
    - Optionally iterate a short fine-tuning phase where τ̂ (threshold approximators) are learned to emulate these quantiles, minimizing L_cov.
- **Implementation details**
    - Mixed precision training (FP16 or BF16)
    - LoRA on selected decoder layers and attention matrices
    - Gradient checkpointing for memory
    - Batch size 1–2 with gradient accumulation to achieve effective batch size 16–32
    - Early stopping based on validation coverage + F1.

Inference strategy

1. **Base generation**
    - Run encoder + decoder with beam search or constrained sampling over structured schema.
    - For each token and field generated, also produce nonconformity scores si.
2. **Prediction set construction**
    - Use stored quantiles qα (or learned τ̂) to define prediction sets:
        - Token-level: include candidates whose si ≤ qα_token.
        - Field-level: aggregate token scores to field score and include field prediction if sfield ≤ qα_field.
3. **Abstention-aware decision**
    - Define a utility function U(F1, coverage, abstention_rate).
    - For each field or document, decide to **accept**, **re-prompt/re-generate**, or **abstain** based on:
        - Whether true coverage constraints are satisfied (approximate)
        - Expected improvement vs. cost of re-generation
    - Implement a simple policy learned on validation set: thresholds on set size and nonconformity statistics.
4. **Output**

    - Final outputs include:
        - Single best structured extraction (for downstream systems)
        - Prediction sets for critical fields (e.g., total amount, date, address)
        - Confidence/coverage estimates at field/document level
        - Flags/abstentions for low-confidence documents.

Uncertainty modeling

- **Type**: Nonparametric distribution-free uncertainty via conformal prediction, not Bayesian approximation.
- **Granularity**: Token-level, field-level, document-level.
- **Signals**: Geometry-aware nonconformity scores combining visual location, layout, and context; not just softmax entropy.
- **Guarantees**: Empirical coverage at or near target levels (e.g., 90–95%) for held-out calibration data; evaluated across document types and corruption levels.
- **Comparison**: Directly compare against MC Dropout, softmax entropy, and auxiliary confidence heads for calibration error (ECE), Brier score, and selective risk.

Evaluation protocol

- **Datasets** (using revised / de-templated versions where available):
    - FUNSD-r (resampled to reduce template leakage)
    - CORD-r / CORD (with template stratification)
    - SROIE-r (re-split to reduce template duplication)
    - DocVQA (single and multipage subsets)
    - Additional real noisy scans and synthetic corruptions (blur, rotation, JPEG artifacts, occlusion).
- **Metrics**:
    - **Accuracy**: F1, EM for field extraction and table reconstruction.
    - **Calibration**: ECE, MCE, Brier score at token, field, and document level.
    - **Coverage & set quality**: Empirical coverage vs target, prediction set size distributions, token/field-level oracle F1 given sets.
    - **Selective prediction**: Risk–coverage curves, AURC, abstention curves (F1 vs. coverage), under both natural and corrupted distributions.
    - **Robustness**: Performance under different noise types/levels and layout shifts.
    - **Efficiency**: Inference time and memory on RTX 5050, comparing confidence methods.

Baselines

1. **Donut-style OCR-free generative model** with:
    - Plain softmax confidence (no uncertainty modeling)
    - MC Dropout-based uncertainty
    - Auxiliary confidence head + calibration loss.
2. **Pix2Struct-style OCR-free model**, similarly augmented.
3. **DocFormerv2-style encoder + standard generative decoder with confidence head** (where feasible within compute).
4. **Conformal wrap baseline**: Post-hoc conformal prediction applied to off-the-shelf Donut/Pix2Struct using simple nonconformity (negative log-probability) without geometry-aware scoring or hierarchical coverage.
5. **Selective prediction baselines** from NLP (calibrated confidence thresholds, entropy-based selection).

Ablations

- **Architecture-level**:
    - With vs without geometry-aware nonconformity (using only sequence logits).
    - With vs without hierarchical (token+field+doc) coverage vs token-only coverage.
- **Loss components**:
    - L_gen only vs L_gen + L_conf vs L_gen + L_conf + L_cov vs full L_total.
    - Vary λ_conf, λ_cov, λ_size to show effect on coverage and set size.
- **Conformal vs classical uncertainty**:
    - Conformal prediction vs MC Dropout vs entropy vs confidence head.
- **Calibration regimes**:
    - Global quantiles vs per-field quantiles vs per-document-type quantiles.
- **Decoding policy**:
    - Greedy decoding vs beam search vs constrained decoding; effect on calibration and coverage.
- **Hardware-aware**:
    - LoRA/adapter configurations and their impact on accuracy and calibration under VRAM constraints.

Expected results

- Comparable or slightly better F1/EM than strong OCR-free baselines on revised CORD/FUNSD/SROIE/DocVQA while:
    - Reducing ECE and Brier score significantly (e.g., 30–50% relative reduction vs MC Dropout/confidence head).

    - Achieving empirical coverage close to targets (e.g., 90–95%) at token, field, document level, even under noise and layout shifts.
    - Producing competitive risk–coverage curves and AURC, clearly outperforming MC Dropout and entropy-based selection.
    - Demonstrating that prediction sets and abstention policies materially improve reliability for high-stakes fields (totals, dates, identifiers).
- Demonstrating **feasible training on RTX 5050** with wall-clock times on the order of a few days per dataset using parameter-efficient tuning.

Why this is novel
Because it **integrates conformal prediction and structured coverage control into the generative decoding process itself**, with geometry-aware nonconformity scoring and hierarchical, differentiable surrogate coverage objectives, rather than treating uncertainty as a post-hoc heuristic add-on. This shifts document extraction from "Donut + MC Dropout" engineering to a **principled, coverage-guaranteed, uncertainty-aware generative framework**, tailored to structured, multimodal document outputs and explicitly engineered to be viable on a single consumer GPU.

Compute feasibility plan

- Use T5-small/base-scale decoder (≤ 220M params) and small ViT/Swin encoder (≤ 100M) with:
    - LoRA on selected attention and feed-forward layers; freeze most backbone parameters.
    - FP16/BF16 training with gradient checkpointing and gradient accumulation.
    - Batch size 1–2, effective batch size 16–32 via accumulation.
    - Dataset sizes limited to standard benchmarks (tens of thousands samples, not pretraining-scale).
    - Conformal calibration computed offline on calibration splits with small batches.
- Estimated:
    - Fine-tuning per dataset: 2–4 days on RTX 5050.
    - Total project: 2–3 weeks active training, within realistic PhD timeline.

Target venues

- Primary: **IEEE TPAMI**, **ACL main conference** (core methodology: structured generative modeling + principled uncertainty).
- Secondary: **NeurIPS/ICLR workshops** on uncertainty and structured prediction; **CVPR/ICCV document understanding workshops** for extended applications.

Final verdict (1 sentence)
This project is a single, coherent, TPAMI/ACL-level research direction: a rigorously calibrated, conformal-prediction-integrated, end-to-end OCR-free generative vision–language architecture for document extraction with genuine algorithmic novelty and realistic single-GPU feasibility.